Short Paper

Is Lexical Semantic Change Measurement-Invariant? Lexicon, Encoder, and Discourse Composition in a Web-Scale Study of ADHD and Autism

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Diachronic measures of lexical semantic change are increasingly used to license claims about cultural change, but those claims inherit whatever the instrument does. We ask whether two such threats—dependence on the resources chosen to operationalise a dimension, and confounding with change in discourse composition—are large enough to alter published conclusions. Building a 13-year corpus of general web discourse from Common Crawl and stratifying 96,864 target mentions by discourse frame with an LLM-in-the-loop classifier, we estimate annual sentiment, intensity, and breadth trajectories for two diagnostic concepts, ADHD and autism, against three negative-emotion comparators, and then re-estimate every trajectory under the alternative lexicon and encoder that the same framework originally specified. Fifteen of twenty-seven series change their descriptive conclusion, and the failures are dimension-specific: sentiment is close to invariant while intensity and breadth are not. Crossing the two lexicons shows why—they agree about valence and disagree about arousal on the same tokens—and a perturbation probe shows the two encoders differ by an order of magnitude in how much non-target material moves them. Substantively, neither predicted form of concept creep is supported, and a shift-share decomposition shows the null is conservative: the corpus composition was drifting in the direction that manufactures apparent creep. We argue that multi-resource comparison and composition stratification belong in the primary analysis of semantic change, not in a robustness appendix.

Keywords: lexical semantic change; diachronic NLP; measurement validity; discourse composition; contextual embeddings; Common Crawl; corpus construction; LLM-assisted annotation; concept creep

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1. Introduction

Computational studies of lexical semantic change (LSC) are increasingly asked to do cultural work. Having established that meaning change can be detected at scale (Hamilton, Leskovec, and Jurafsky 2016; Schlechtweg et al. 2020), the field is now routinely used to argue that a society's concepts of harm, illness, or identity have shifted (Vylomova and Haslam 2021; Baes, Haslam, and Vylomova 2023). That is a considerable inferential burden for a measurement to carry, and it makes the properties of the measurement itself a first-order question. A trend estimated from lexicon-weighted collocates or embedding dispersion is only evidence about a concept if it is stable under choices that were never meant to be substantive.

Two such threats are well known individually and rarely tested together. The first is *operationalisation dependence*. LSC estimates have repeatedly proved sensitive to things that ought not to matter: the random seed and corpus slice used to fit embeddings (Antoniak and Mimno 2018), the frequency and polysemy profile of the target against a proper control (Dubossarsky, Weinshall, and Grossman 2017), and even orthographic preprocessing before a contextual encoder (Laicher et al. 2021). The second is *discourse-composition confounding*. A word's measured context can change because its meaning changed, or because the population of texts that use it changed. Working on newspaper text, Pisl, Bucur, and Podina (2025) showed that an apparent semantic-intensity trend for *depression* became non-significant once the balance of clinical and non-clinical contexts was controlled.

We test both in a single design, on a case that carries a strong standing directional prediction. Concept creep theory holds that harm-related concepts broaden to admit qualitatively new instances (horizontal creep) and loosen to admit milder ones (vertical creep) (Haslam 2016; Haslam et al. 2020), and a series of studies reports exactly that for mental-health vocabulary (Vylomova and Haslam 2021; Baes et al. 2023; Baes, Haslam, and Vylomova 2023). ADHD and autism are useful test cases: they are unambiguously diagnostic concepts, so the polysemy that complicates *depression* or *anxiety* does not arise (Xiao et al. 2023), and they sit at the centre of contemporary argument about whether the popularisation of clinical language erodes its meaning (Isern-Mas and Almagro 2025; Spencer and Carel 2021). If a directional prediction of this strength cannot be evaluated invariantly, weaker ones cannot either.

The paper makes five contributions.

1. A reusable pipeline for constructing diachronic corpora of general web discourse from Common Crawl, using a WET-first scan with WARC-validated extraction, applied here to 13 annual snapshots from 2014 to 2026 (Section 3).
2. A frame-aware design that separates discourse composition from meaning before trends are estimated, with frame labels produced by an LLM annotator under critical review and human adjudication (LLM–human κ between .87 and .95) and audited for temporal stability (Section 4).
3. Evidence that the intensity and breadth dimensions of a published LSC framework are not measurement-invariant, while its sentiment dimension largely is: substituting the framework's own original lexicon and encoder changes the descriptive conclusion for 15 of 27 series, concentrated in two of the three dimensions. This holds for the comparator terms as well as the targets, so it is not an artefact of the case (Section 5.5).

4. An account of *why* each substitution matters: crossing the two lexicons attributes their disagreement to ratings rather than coverage, and a perturbation probe shows the two encoders differ by an order of magnitude in how much non-target material moves their representations (Section 5.6).
5. A bounded null result for concept creep in two institutionally codified diagnostic concepts, with the smallest detectable linear changes stated explicitly (Sections 5.2–5.3).

Code, configuration, and the derived annual estimates behind every figure and table are available at `https://github.com/jako6f/msc-nlp-therapy-speak`.

2. Background

2.1 Measuring Lexical Semantic Change

Computational LSC detection has moved through three broad representations (for surveys see Tahmasebi, Borin, and Jatowt 2019; Tang 2018; Kutuzov et al. 2018; Periti and Montanelli 2024; de Sá, Silveira, and Pruski 2026). Type-level embeddings assign one vector per word per period and compare them across aligned spaces (Hamilton, Leskovec, and Jurafsky 2016; Mikolov et al. 2013). Contextual embeddings assign a vector per usage, so change can be characterised as a shift in the distribution of usages rather than in a single point (Giulianelli, Del Tredici, and Fernández 2020). Target-aware word-in-context models go further and encode the target occurrence specifically, which is what the word-in-context task requires (Pilehvar and Camacho-Collados 2019); XL-LEXEME is trained for exactly this and is the current default for LSC detection (Cassotti et al. 2023).

Alongside this progress runs a persistent methodological worry. Embedding-derived similarity is unstable across corpus samples and initialisations (Antoniak and Mimno 2018); apparent "laws" of semantic change partly reflect frequency and polysemy artefacts that only proper controls reveal (Dubossarsky, Weinshall, and Grossman 2017); results differ across corpora and time spans for the same targets (Schlechtweg et al. 2019); and even preprocessing decisions upstream of a contextual encoder move measured change substantially (Laicher et al. 2021).

These results share a structure worth making explicit, because it shapes what we test here. Each identifies a degree of freedom that the analyst does not experience as a substantive choice—a seed, a sampling window, a normalisation step—and shows that it moves the estimate. But each also concerns the *detection* stage: whether a word changed, and by how much. The frameworks now being used to make cultural arguments add a second stage on top of detection, in which change is decomposed into interpretable dimensions and each dimension is estimated from a designated resource—an affect lexicon, an encoder. That second stage introduces degrees of freedom of its own, and they have not been probed in the same way. Two questions follow, and we take both. Does substituting a resource that the framework's own authors treat as interchangeable reverse a substantive finding? And if so, can the mechanism be identified well enough that a practitioner using a single resource could detect their own exposure to it?

2.2 Characterising Change: The SIBling Framework

Most LSC work asks *whether* a word changed. Baes, Haslam, and Vylomova (2024) ask *how*, proposing the SIBling framework—Sentiment, Intensity, Breadth—to characterise change along interpretable dimensions rather than as one aggregate distance. Sentiment and intensity are estimated as the mean valence and arousal of affect-rated collocates in a local window; breadth is the mean pairwise distance among contextual embeddings of the target's usages. The three dimensions map onto the LSC vocabulary of amelioration and pejoration, of strengthening and weakening, and of narrowing and broadening (Campbell 2013), and the framework is explicitly offered to social scientists as a way to make cultural inferences from lexical evidence.

The authors describe their operationalisations as first implementations that have not been externally validated and remain open to refinement. This paper takes that caveat as a research question. Two substitutions are natural. For affect, the Warriner norms (Warriner, Kuperman, and Brysbaert 2013) used in the original can be replaced by NRC–VAD v2.1 (Mohammad 2025), which offers roughly threefold lexical coverage, multi-word entries, and higher reported reliability. For breadth, the generic sentence encoder (Song et al. 2020; Reimers and Gurevych 2019) can be replaced by a target-aware model (Cassotti et al. 2023), on the argument that the analytic object is a term's usage rather than a document's topic. Both substitutions are defensible *a priori*. The empirical question is whether choosing between them changes what one concludes.

2.3 The Hypothesis Under Test

Concept creep theory predicts that harm-related concepts loosen vertically and broaden horizontally over time (Haslam 2016, 2026). Concepts of mental illness qualify as harm-related because distress and dysfunction are central to their definition (Wakefield 1992). The evidence base is substantial but mixed. Creep has been reported for *trauma*, *addiction*, *bullying*, *prejudice*, and *harassment* in psychology abstracts and in curated general-language corpora (Vylomova and Haslam 2021; Baes, Haslam, and Vylomova 2023). Against this, Xiao et al. (2023) found the semantic intensity of *anxiety* and *depression increasing*, and Iacob and Uban (2026) found most therapy-speak terms broadening in psychology abstracts but narrowing on Reddit, with long-established clinical terms barely moving at all.

Two features of this record motivate our design. First, it rests almost entirely on curated corpora, whose composition reflects editorial and disciplinary conventions as much as public usage; general web discourse has not been examined systematically, despite being the substrate from which large-scale corpora are now routinely built (Wenzek et al. 2019; Penedo et al. 2024). Second, the divergent findings are consistent with both threats we test: they could reflect real differences between concepts, or shifting composition, or instrumentation. The compositional threat has a specific and directional form here. Clinical contexts are affectively more intense than lived-experience ones, so a corpus whose mixture drifts toward clinical language will show rising measured intensity and one drifting away from it will show falling intensity, whichever way the meaning of the term is moving. A drift of the first kind is what this literature calls *pathologisation*, and measures of it have been built directly into the collocate-based tradition (Baes, Haslam, and Vylomova 2023, 2024). An aggregate estimate conflates pathologisation with semantic change unless the mixture is modelled, which is precisely the confound Pisl, Bucur, and Podina (2025) identify for *depression* in newspaper text.

We therefore evaluate the two directional creep hypotheses—declining intensity and increasing breadth for ADHD and autism—as the substantive test, and treat sentiment and frame composition as exploratory.

3. Corpus Construction

3.1 Common Crawl as an Evidentiary Base

Common Crawl is the largest openly available web archive, published roughly monthly with more than two billion pages per crawl (Common Crawl Team 2024). Its value here is breadth: it provides repeated cross-sections of general web discourse rather than one platform or newspaper archive.

Its limits should be stated before its results, not after. Common Crawl is neither a complete copy of the web nor a representative sample of it (Baack 2024). Since 2017 domain selection has followed harmonic centrality, so well-linked domains are both likelier to be crawled and more deeply crawled; large social platforms are absent; and rights-holder blocking via `robots.txt` has grown over precisely the years we study. Temporal comparability is therefore imperfect by construction. Following the spirit of data statements (Bender and Friedman 2018; Elazar et al. 2024), we take the corpus to license claims about quality-filtered, English-language open-web discourse as archived by Common Crawl, and about nothing wider.

3.2 Pipeline

Figure 1 shows the collection design. It exploits the two Common Crawl formats sequentially. WET files hold compact extracted plaintext and are cheap to scan, so they carry large-scale term matching; but they lack the HTML structure needed for boilerplate removal and metadata recovery. Candidate documents are therefore resolved to their WARC records, where the original HTML is fetched by byte range and main text re-extracted. Reserving WARC processing for candidates only is what makes web-scale diachronic collection affordable: the full run scanned 220 million pages on a single two-vCPU cloud instance at roughly one hour per million WET records.

Software choices affect corpus membership and are therefore reported. WET and WARC records are parsed with `warcio`; WARC pointers resolve through a local `pywb` index server; archived HTML is converted to main text with Trafilatura and Resiliparse (Barbaresi 2021; Bevendorff et al. 2018); publication dates are recovered with `htmldate` (Barbaresi 2020); and post-extraction filtering applies DataTrove quality filters (Penedo et al. 2026) followed by language identification with `py3langid` (Lui and Baldwin 2012). One crawl is selected per year near a fixed anchor date and frozen in a crawl map, so the sampling frame is deterministic and re-runnable. To limit domination by large sites beyond Common Crawl's own sampling, we cap retained hits at 50 rows per registered domain per crawl.

Across 2014–2026 the collection scanned 220 million pages and retained 336,178 analysis-ready documents, of which 87,173 contain an ADHD or autism term.

3.3 Targets, Comparators, and Contexts

The targets are matched with frozen expressions covering `adhd` and `attention[-\s]?deficit` for ADHD, and `autism`, `autistic`, `autism[-\s]?spectrum`, and `ASD` for autism, with `ASD` retained only when *autism* occurs within

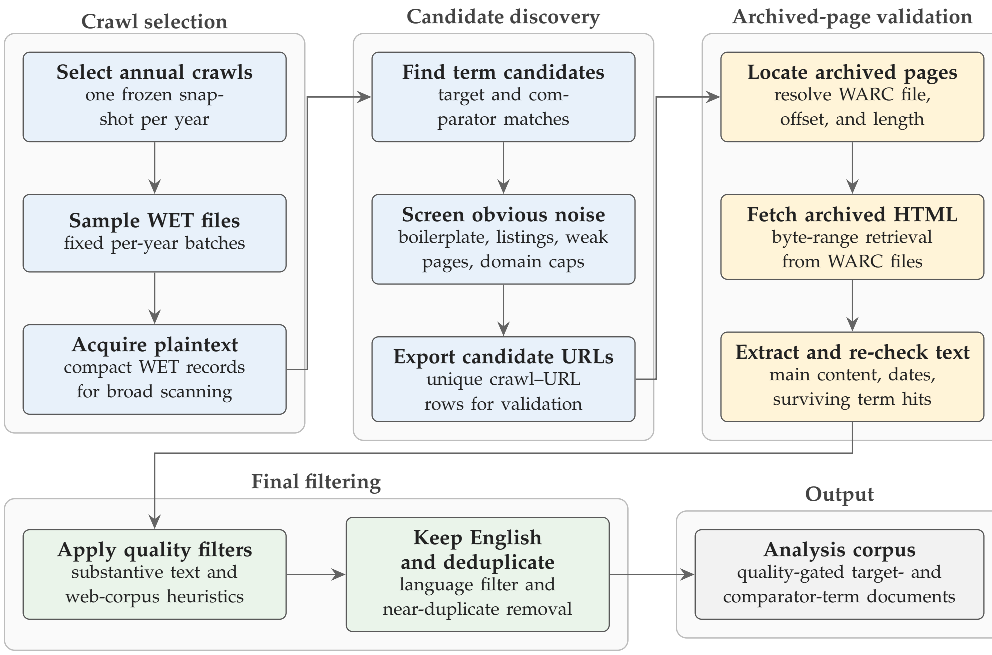

Figure 1
The collection pipeline. *Crawl selection* freezes one snapshot per year and acquires its compact WET plaintext. *Candidate discovery* scans that plaintext for target and comparator matches, screens obvious noise, and exports candidate URLs. *Archived-page validation* resolves each candidate to its WARC record, fetches the original HTML by byte range, and re-extracts main text, dates, and surviving term hits. *Final filtering* applies document-quality gates, an English filter, and near-duplicate removal, yielding the quality-gated analysis corpus. Reserving the expensive WARC stage for candidates only is what makes web-scale diachronic collection affordable.

±200 characters. Three negative, non-clinical emotion terms—*frustration*, *sadness*, *loneliness*—serve as comparators. They are not matched semantic controls in the sense of Dubossarsky, Weinshall, and Grossman (2017); they establish whether the instruments detect movement at all in this corpus over this window, which is what a null result on the targets requires.

The unit of analysis is the individual mention rather than the document. Overlapping matches are resolved to the longest span, and same-sentence acronym–expansion pairs are collapsed so that "attention deficit hyperactivity disorder (ADHD)" contributes one context rather than two; each document may contribute at most three mentions per analysis unit, which limits domination by repetitive documents while preserving repeated-use signal.

The attrition from documents to mentions is worth reporting, because in web corpora it is where silent selection happens. The 336,178 analysis-ready documents yielded 664,606 raw matches; overlap resolution reduced this to 643,735, acronym–expansion collapse to 633,026, and the three-mention cap to 465,400 candidate contexts. The publication-year filter then removed 161,122 contexts published before 2014 and 10,608 with no parseable date. The large document-level attrition, from 336,178 to 212,651, therefore reflects mostly archived pages published before the study window rather than aggressive filtering: each yearly crawl also captures older pages. The result is 293,670 mention contexts from 212,651 documents across 135,771 registered domains, including 96,864 target contexts (28,611 ADHD, 68,253 autism).

Each context stores a ± 5-token window for affective analysis and the target sentence with adjacent context for embedding analysis. Documents may contain both targets—11,795 do—and contribute separately to each, which matters for interpretation: as the two concepts are increasingly discussed together, each becomes part of the other's measured context (Kang, Haslam, and Conway 2025).

4. Frame-Aware Design and Measures

4.1 Frame Classification

In web text, ADHD and autism appear as diagnoses, disorder constructs, service categories, identities, community labels, and incidental boilerplate. Treating all mentions as one semantic population risks reading a change in that mixture as a change in meaning. We therefore label each target passage—the target sentence plus adjacent context—before estimating any trend.

Labelling is hierarchical. A passage is first coded for whether it contains substantive target discourse; passages that are thin, navigational, promotional, or otherwise insufficient for target-specific interpretation are set aside. Substantive passages are then coded on two non-exclusive axes: whether clinical framing is present (diagnosis, disorder status, symptoms, impairment, treatment, services, medication, research, epidemiology, DSM/ICD-style categories, and educational or clinical support needs) and whether lived-experience framing is present (identity, self-understanding, family or first-person experience, neurodivergent community, stigma, accommodation, everyday coping, belonging, pride, and embodied or social experience).

The axes are deliberately non-exclusive, because a passage can do both—a first-person account of receiving a diagnosis is simultaneously clinical and experiential—and forcing a single choice would push exactly the ambiguous cases into arbitrary bins. They combine deterministically into five derived strata: clinical but not lived (*clinical-only*), lived but not clinical (*lived-only*), both (*mixed*), substantive but neither (*substantive-other*), and non-substantive. The analyses report the two unambiguous strata alongside the substantive aggregate. Mixed contexts are estimated but not plotted separately: this smaller stratum generally tracked between the overall and clinical trajectories and added visual overlap without changing any conclusion.

Labels were produced by an Annotation with Critical Thinking workflow (Lin et al. 2025). A 200-passage human pilot fixed the codebook. The locked codebook was then applied to 3,000 passages, stratified by year and target, by a frozen annotator (`gpt-5.5`, accessed through the Codex CLI at high reasoning effort with web search disabled); a second, independent model (`gemini-3-flash-preview`, accessed through the Gemini CLI) reviewed every annotation and assigned an error-risk score; and a human coder reviewed and corrected the 400 highest-risk cases plus a random sample of low-risk cases. No model suggestion changed a label without human review. This follows the emerging consensus that LLM annotation is useful but requires validation against human judgement rather than substituting for it (Gilardi, Alizadeh, and Kubli 2023; Pangakis, Wolken, and Fasching 2023; Ziems et al. 2024).

The corrected labels trained a hierarchical classifier over sentence embeddings (Song et al. 2020): three balanced logistic heads, one predicting substantive discourse over all labelled examples and two predicting the frame axes among substantive ex-

Table 1
Annotation reliability and held-out classifier validation for the three hierarchical frame decisions. LLM–human agreement compares the locked Codex annotator against the human coder on the 200-passage pilot; classifier performance is measured on the disjoint 200-passage validation set, which was withheld from codebook development, LLM annotation, criticism, correction, and training.

	LLM–human agreement			Held-out classifier			
Decision	n	Agr.	κ	Support	P	R	F1
Substantive target discourse	200	.960	.914	141/200	.887	.837	.861
Clinical framing	122	.943	.867	105/141	.903	.886	.894
Lived-experience framing	122	.975	.949	46/141	.704	.826	.760

Note. The clinical and lived-experience rows are evaluated among passages the human coder judged substantive. All human labels come from a single coder, so no inter-annotator agreement is reportable.

amples. Year, URL, and domain features are excluded to avoid leakage from temporal or source artefacts. A separate 200-passage validation set was withheld from codebook development, LLM annotation, criticism, correction, and training.

Table 1 reports both halves of this instrument. The LLM annotator agrees with the human coder closely on all three decisions: $\kappa = .914$ on whether a passage is substantive target discourse (96.0% raw agreement over 200 passages), $\kappa = .867$ on clinical framing, and $\kappa = .949$ on lived-experience framing (both over the 122 passages the coder judged substantive). Agreement on the five-way derived stratum, which requires all three decisions to coincide, is $\kappa = .877$. These are high enough that the LLM stage is doing recognisably the same task as the human one, which is what licenses using it at a scale hand-coding could not reach.

The classifier trained on those labels transfers well to the held-out set. Substantive-discourse detection reaches F1 = .861 and clinical framing F1 = .894, both close to the agreement ceiling above. Lived-experience framing is the weakest head (F1 = .760, precision = .704), which matters for interpretation: lived-experience strata contain clinical admixture, so frame contrasts are attenuated and the frame-specific trends reported below are, if anything, conservative. One caveat belongs with all of these numbers: every human label in this study comes from a single coder, so no inter-annotator agreement is reportable and validation is measured against an individual rather than a consensus standard.

A post-hoc diagnostic checks the frame distinction without involving the classifier at all. Fitting annual skip-gram models separately within each frame and extracting each target's recurrent nearest neighbours (Mikolov et al. 2013; Vylomova and Haslam 2021) recovers the same division from distributional evidence alone (Appendix F). For autism the two neighbour sets are almost disjoint: clinical contexts yield *developmental*, *disorder*, *research*, and *study*, lived-experience contexts *family*, *life*, *people*, and *support*. ADHD shows the same contrast more weakly, with *disorder*, *symptom*, and *condition* against *help*. The frames therefore track a real difference in how these terms are used, not an artefact of the codebook.

4.2 Measures

For analysis unit u, publication year Y, and stratum s, **sentiment** and **intensity** are the frequency-weighted mean valence $V(w)$ and arousal $A(w)$ of affect-rated collocates in the ± 5-token windows,

$$\text{Sentiment}_{u,Y,s} = \frac{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}\, V(w)}{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}}, \quad \text{Intensity}_{u,Y,s} = \frac{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}\, A(w)}{\sum_{w \in C_{u,Y,s}} f_{w,u,Y,s}}, \tag{1}$$

where $C_{u,Y,s}$ is the set of matched collocates and f their frequencies. The focal term is removed from its own window. Windows are tokenised, tagged, and lemmatised with spaCy (Honnibal et al. 2020); punctuation, symbols, numerals, one-character lemmas, and stopwords are excluded; lexicon entries are normalised identically, and multi-word entries matched greedily before unigrams.

Breadth operationalises horizontal creep as contextual dispersion. Each context is marked with explicit target delimiters and encoded by XL-LEXEME, and breadth is the mean pairwise cosine distance among the $N_{u,Y,s}$ resulting L2-normalised target-use embeddings,

$$\text{Breadth}_{u,Y,s} = \frac{2}{N_{u,Y,s}(N_{u,Y,s}-1)} \sum_{i<j} \left(1 - \mathbf{v}_i \cdot \mathbf{v}_j\right). \tag{2}$$

Higher values mean more heterogeneous usage. The implementation evaluates this in closed form over the normalised vectors rather than materialising the pairwise matrix.

4.3 Alternative Operationalisations

Every affective and breadth trajectory is estimated twice. The primary estimates use NRC–VAD and XL-LEXEME; the alternative estimates use the resources SIBling originally specified, the Warriner norms and MPNet sentence embeddings, with the annual strata, sampling, and trend-model conventions held constant.

One constraint governs how the two can be compared. The lexicons live on different native scales—NRC–VAD spans -1 to 1, the Warriner norms are a 1–9 scale rescaled to 0–1—so raw slope magnitudes are not commensurable. We therefore compare only the *sign*, the *significance verdict*, and the *standardised* year coefficient, never the unstandardised slope. The MPNet breadth track is extended to the comparator terms here, so every dimension is compared over the same set of series.

4.4 Statistical Analysis

Annual estimates are the objects of interpretation. Uncertainty intervals come from document-level bootstrap resampling within each unit–year–stratum cell (500 repetitions, 2.5th and 97.5th percentiles); the document is the resampling unit because mentions and collocates from one document are not independent. Ordinary least squares over the 13 annual points summarises net linear movement. Residual autocorrelation is assessed with a Durbin–Watson statistic and flagged outside $[1.25, 2.75]$; flagged series are re-estimated with a first-order autoregressive model reported alongside (Chatterjee and Hadi 2012). Quadratic fits are computed only for flagged series and treated as diagnostics, not as competing trend estimates.

Two constraints on interpretation follow, and both are load-bearing for a paper whose substantive result is a null. First, all p-values are uncorrected. The primary operationalisation contributes 34 trend tests and the alternative a further 27, so at $\alpha = .05$ roughly three nominally significant slopes are expected by chance across the

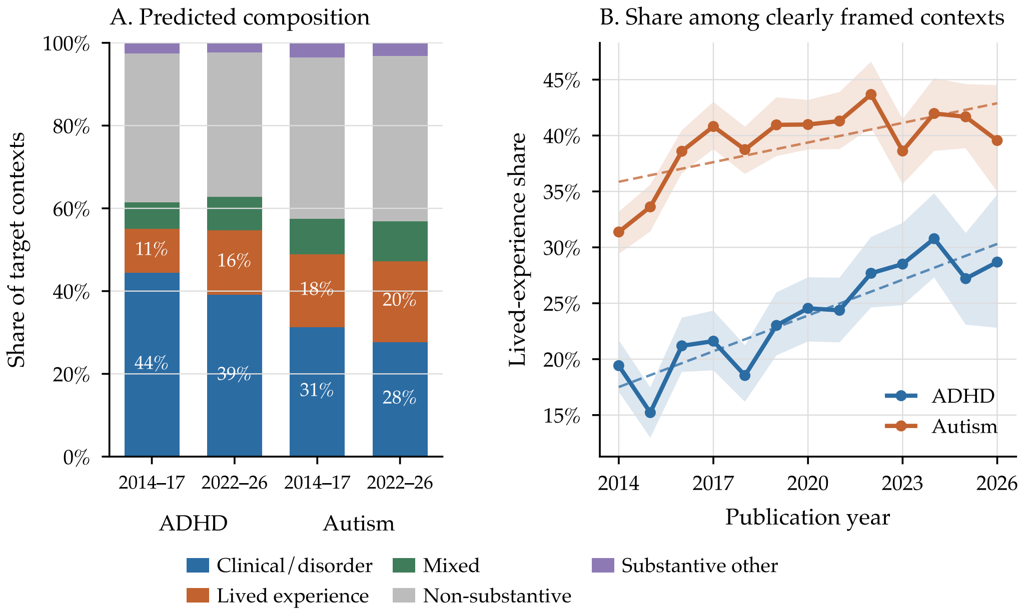

Figure 2
Frame composition of target contexts. Panel A compares the full predicted composition in the early and late periods. Panel B gives the annual lived-experience share among contexts assigned one clear frame, with 95% document-bootstrap bands and OLS trend summaries.

comparison; isolated significant secondary slopes should be read accordingly, and we mark them as nominal throughout. Second, with 13 annual observations a slope reaches $p < .05$ only when $|B| \geq 2.20 \times SE$. For the four directional hypothesis tests, the fitted standard errors imply that the smallest 2014–2026 changes that would have been detected are about .013 (ADHD) and .022 (autism) on the arousal index, and .008 and .006 in mean pairwise cosine distance. The null results below rule out linear changes larger than these; they do not rule out smaller drifts.

5. Results

5.1 Discourse Composition Shifts

Across all 96,864 labelled contexts, clinical-only framing accounts for 42.1% of ADHD and 29.8% of autism contexts and lived-only for 12.6% and 18.8%; the remainder are mixed (7.0%, 9.3%), non-substantive (35.8%, 38.8%), and substantive-other (2.5%, 3.3%).

The composition moved (Figure 2, Table 2). Among contexts assigned one clear frame, the lived-experience share of ADHD discourse rose by 1.07 percentage points per year ($p < .001$, adjusted $R^2 = .80$), a fitted increase from 17.5% to 30.3%. Autism moved the same way from a much higher base, 35.9% to 42.9% ($B = 0.59$, $p = .013$), but this series was flagged for residual autocorrelation and the AR(1) sensitivity slope did not differ from zero ($p = .370$); we therefore treat the autism compositional trend as unresolved and the ADHD one as robust.

Table 2
Annual trend models for the lived-experience share of clearly framed target contexts, defined as lived-only divided by lived-only plus clinical-only.

Target	B (SE), pp/yr	β	Adj. R^2	Fitted 2014	Fitted 2026	AR(1) B (p)
ADHD	1.07*** (0.15)	+.90	.80	17.5%	30.3%	—
Autism	0.59*† (0.20)	+.67	.39	35.9%	42.9%	.23 (p = .370)

Note. B is the annual OLS slope in percentage points; β is the standardised year coefficient. † marks a residual-autocorrelation flag, for which the AR(1) sensitivity slope is reported in the final column. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$; p-values are descriptive and uncorrected.

Because this trend is measured through a classifier, it is exposed to the possibility that classifier error drifted over time. A period-stratified audit (Appendix A) finds the opposite of what that explanation requires: lived-experience precision is *highest* in the late period (.792, against .667 early), and the human-adjudicated labels reproduce the rise independently of the classifier (+1.11 percentage points per year, bootstrap 95% CI [0.40, 1.88]). With only 11–23 lived-experience validation positives per period, modest error drift cannot be excluded and the fitted 12.8-point rise may overstate the true change.

5.2 No Vertical Creep

Intensity estimates rest on 17,649 ADHD and 39,463 autism contexts, of which 97.8% and 99.3% contributed at least one lexicon match, yielding 58,666 and 144,668 matched collocate occurrences.

Vertical creep predicts declining arousal. It declined in no target stratum (Table 3, Figure 3). For ADHD the slope did not differ from zero overall ($B = 0.0001$, $p = .900$), in clinical contexts ($p = .738$), or in lived-experience contexts ($p = .123$). For autism the overall slope was positive but not distinguishable from zero ($p = .223$); the only slope that reached significance was a nominal *increase* in clinical contexts ($B = 0.0010$, $p = .048$), which is within the range expected by chance across 34 tests and which we do not interpret. Given the detectable-effect bound in Section 4.4, the finding is that arousal did not fall by more than about .013 (ADHD) or .022 (autism) index units over thirteen years.

5.3 No Horizontal Creep—and Contraction

Breadth estimates rest on 57,112 target contexts in the three substantive frames, 53,544 after within-cell duplicate removal, with 38,349 comparator rows.

Horizontal creep predicts increasing dispersion. Where breadth moved reliably, it moved the other way. ADHD contracted overall, from .138 in 2014 to .117 in 2026 ($B = -0.0020$, $p < .001$, adjusted $R^2 = .77$), and in clinical contexts ($B = -0.0018$, $p < .001$); these are the strongest semantic trends in the study. Autism lived-experience breadth also declined ($B = -0.0010$, $p = .030$). Autism overall was flat ($p = .637$), and the nominally positive autism clinical slope ($p = .036$) was not retained by the AR(1) model ($p = .755$).

A post-hoc diagnostic ranks contexts by their average distance to others in the same cell (Appendix E). The most distant ADHD contexts remain saturated with diagnostic and educational vocabulary—*disorder*, *impulsivity*, *inattention*, *learning*, *dyslexia*—rather

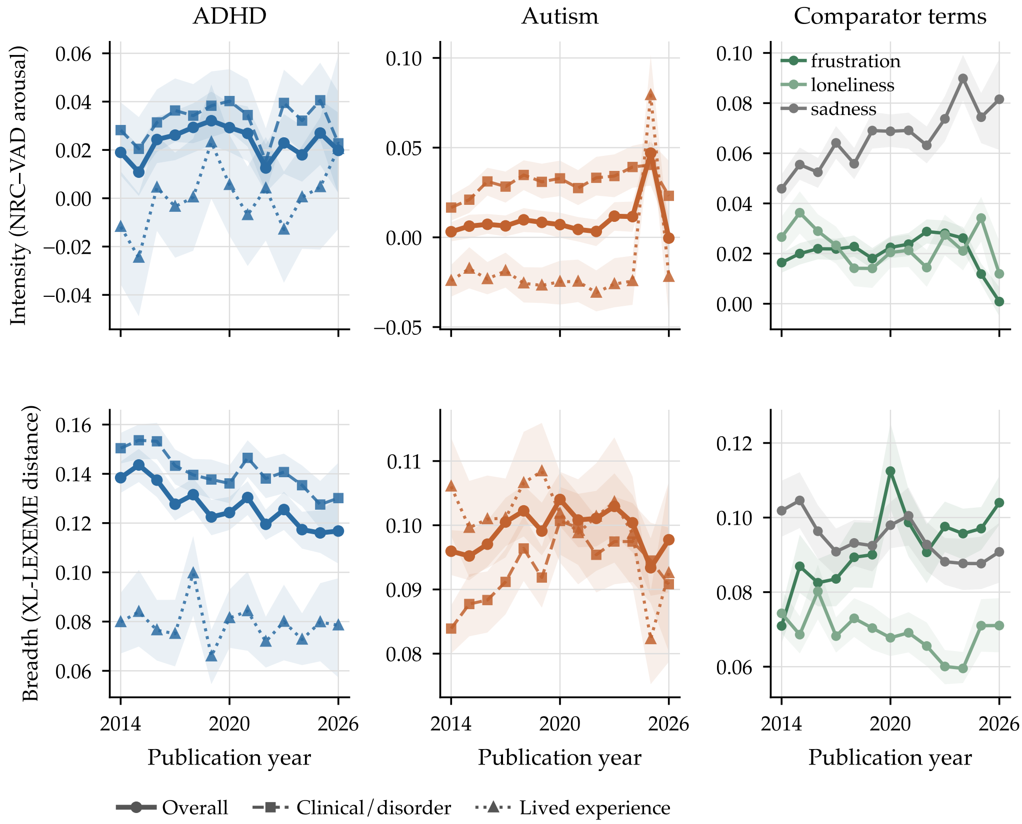

Figure 3
Annual intensity (top) and breadth (bottom) for the two targets by frame and for the three comparator terms, with 95% document-bootstrap bands. Neither predicted form of concept creep appears: intensity does not decline, and breadth contracts rather than broadens. The 2026 estimates rest on the thinnest annual samples, because each crawl captures few pages published in its own year.

than reaching into new domains. That the comparator *frustration* broadened over the same window ($B = 0.0020$, $p < .01$) indicates the contraction is target-specific rather than a corpus-wide artefact.

5.4 Sentiment

No target sentiment slope differed from zero in any stratum under the primary lexicon (Table 3). The autocorrelation-flagged series traced U-shapes with minima between Q3 2018 and Q2 2019, and the AR(1) model gave a positive valence trend in autism clinical contexts ($B = 0.0052$, $p = .040$). Quadratic coefficients are in Appendix D. We note the turning point but do not build on it: it rests on diagnostics run because linear residuals were flagged, not on a pre-specified test.

Table 3
Annual trend models for the three semantic measures, by target and frame. Sentiment and intensity are NRC–VAD valence and arousal over local collocates; breadth is the mean pairwise cosine distance among XL-LEXEME target-use embeddings.

Measure	Target	Frame	B (SE)	β	Adj. R^2
Sentiment	ADHD	Overall	.0023† (.0014)	+.46	.14
	ADHD	Clinical	.0013† (.0015)	+.24	−.03
	ADHD	Lived experience	.0002 (.0017)	+.03	−.09
	Autism	Overall	.0012 (.0006)	+.50	.18
	Autism	Clinical	.0022† (.0015)	+.42	.10
	Autism	Lived experience	−.0038 (.0022)	−.46	.14
Intensity	ADHD	Overall	.0001 (.0005)	+.04	−.09
	ADHD	Clinical	.0002 (.0006)	+.10	−.08
	ADHD	Lived experience	.0015 (.0009)	+.45	.13
	Autism	Overall	.0011 (.0008)	+.36	.05
	Autism	Clinical	.0010* (.0004)	+.56	.25
	Autism	Lived experience	.0025 (.0021)	+.34	.04
Breadth	ADHD	Overall	−.0020*** (.0003)	−.89	.77
	ADHD	Clinical	−.0018*** (.0003)	−.85	.69
	ADHD	Lived experience	−.0004† (.0006)	−.18	−.06
	Autism	Overall	.0001† (.0002)	+.14	−.07
	Autism	Clinical	.0007*† (.0003)	+.58	.28
	Autism	Lived experience	−.0010* (.0004)	−.60	.30

Note. B is the unstandardised annual OLS slope over 13 publication years; β is the standardised year coefficient. † marks a residual-autocorrelation flag; AR(1) sensitivity estimates for flagged series are reported in Appendix C. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$; p-values are descriptive and uncorrected across 34 trend tests.

5.5 Operationalisation Sensitivity

We now re-estimate every series with the resources SIBling originally specified. Table 4 asks, for each series, whether the two resources license the same descriptive conclusion—the same significance verdict and, where significant, the same sign. Fifteen of twenty-seven do not.

The disagreement is concentrated by dimension rather than distributed across them. Sentiment is close to invariant: eight of nine series agree, the exception being autism overall, which sits either side of the threshold ($p = .081$ under NRC–VAD, $p = .018$ under Warriner). Intensity fails in six of nine, and breadth in eight of nine. The two dimensions that carry the concept-creep hypotheses are the two that do not survive substitution, while the dimension we treated as exploratory is the one that does.

The individual disagreements are large. Under Warriner norms, ADHD intensity declines reliably overall ($p = .001$) and in clinical contexts ($p = .002$) where NRC–VAD found nothing—a result that, reported alone, would have been presented as textbook vertical concept creep. The nominal autism clinical increase disappears ($p = .203$). Under MPNet, the pronounced XL-LEXEME contraction in ADHD breadth attenuates to a null ($p = .445$) and in ADHD clinical contexts vanishes entirely ($\beta = -0.01$, $p = .980$), while autism overall breadth, flat under XL-LEXEME, becomes a reliable decline ($p = .001$).

The comparators rule out a target-specific explanation. If only the targets flipped, one could argue these particular concepts are hard to measure. But *sadness*, whose intensity rose under NRC–VAD with the largest standardised coefficient in the comparator set ($\beta = +0.88$, $p < .001$), shows nothing under Warriner ($\beta = +0.23$, $p = .449$); *frustra-*

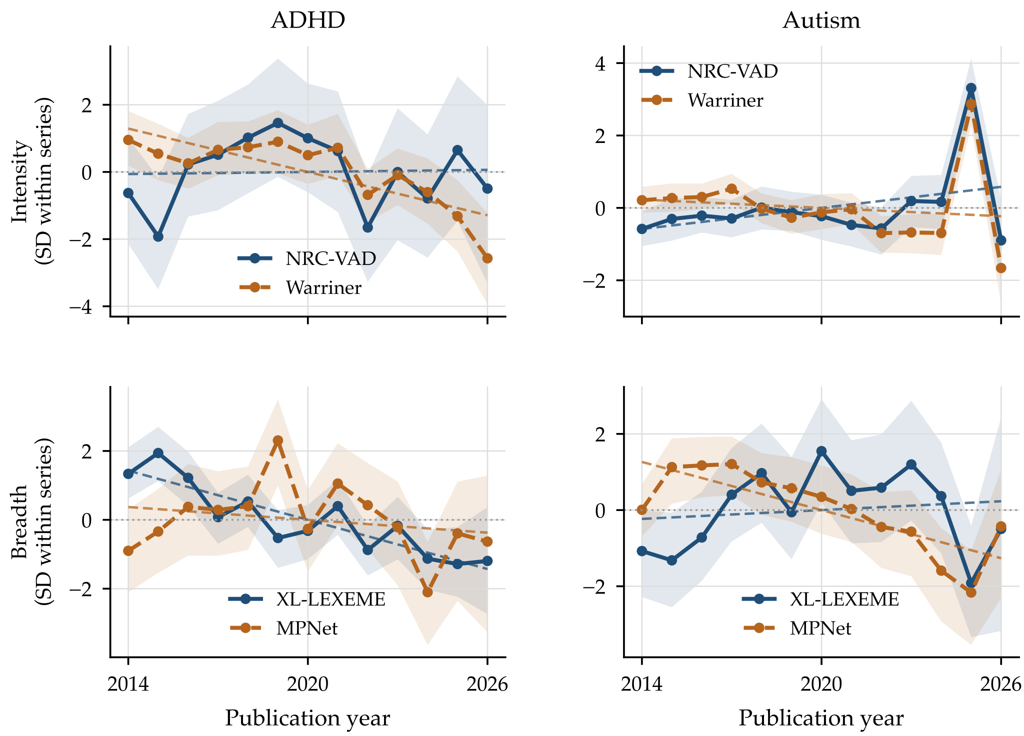

Figure 4
The same series estimated under the primary and the original operationalisation. Because the two affective lexicons use different native scales, each series is standardised by its own across-year standard deviation; standardisation is a linear transform, so signs, significance, and standardised slopes are exactly as reported in the trend tables. Dashed lines are OLS fits.

tion and *loneliness*, null under NRC–VAD, both decline reliably under Warriner; and for breadth all three comparators disagree across encoders, with *frustration* reversing from reliable broadening under XL-LEXEME to a non-significant contraction under MPNet. The instruments reverse conclusions about ordinary emotion vocabulary as readily as about diagnostic vocabulary.

One conclusion survives both operationalisations for both targets and can be stated with corresponding confidence: *neither target broadened horizontally*. Every reliable breadth movement observed under either encoder is a contraction or a null. The vertical-creep conclusion is measurement-conditional, and the sentiment conclusion holds for every series except autism overall.

5.6 Where the Disagreement Comes From

Identifying the mechanism indicates which studies are at risk. For the lexicons it can be isolated arithmetically; for the encoders it cannot, so we probe them behaviourally.

5.6.1 Lexicons: ratings, not coverage. Both lexicons were applied to the same collocate windows, so their matches share a context index and can be crossed. Scoring the shared

Table 4
Conclusion concordance across operationalisations. Each row asks whether the two resources license the same descriptive conclusion for the same series: agreement requires the same significance verdict at $p < .05$ and, where significant, the same sign.

Series	Frame	Primary β	Alternative β	Conclusion
Sentiment: NRC–VAD versus Warriner				
ADHD	Overall	$+.46$	$+.27$	agree
ADHD	Clinical	$+.24$	$-.14$	agree
ADHD	Lived experience	$+.03$	$-.19$	agree
Autism	Overall	$+.50$	$+.64^{*}$	**differ**
Autism	Clinical	$+.42$	$+.51$	agree
Autism	Lived experience	$-.46$	$-.30$	agree
frustration	Baseline	$+.67^{*}$	$+.64^{*}$	agree
loneliness	Baseline	$+.71^{**}$	$+.72^{**}$	agree
sadness	Baseline	$-.79^{**}$	$-.84^{***}$	agree
Intensity: NRC–VAD versus Warriner				
ADHD	Overall	$+.04$	$-.80^{***}$	**differ**
ADHD	Clinical	$+.10$	$-.78^{**}$	**differ**
ADHD	Lived experience	$+.45$	$-.33$	agree
Autism	Overall	$+.36$	$-.14$	agree
Autism	Clinical	$+.56^{*}$	$-.38$	**differ**
Autism	Lived experience	$+.34$	$+.12$	agree
frustration	Baseline	$-.23$	$-.65^{*}$	**differ**
loneliness	Baseline	$-.30$	$-.73^{**}$	**differ**
sadness	Baseline	$+.88^{***}$	$+.23$	**differ**
Breadth: XL-LEXEME versus MPNet				
ADHD	Overall	$-.89^{***}$	$-.23$	**differ**
ADHD	Clinical	$-.85^{***}$	$-.01$	**differ**
ADHD	Lived experience	$-.18$	$-.63^{*}$	**differ**
Autism	Overall	$+.14$	$-.79^{**}$	**differ**
Autism	Clinical	$+.58^{*}$	$-.09$	**differ**
Autism	Lived experience	$-.60^{*}$	$-.61^{*}$	agree
frustration	Baseline	$+.72^{**}$	$-.53$	**differ**
loneliness	Baseline	$-.51$	$-.81^{***}$	**differ**
sadness	Baseline	$-.71^{**}$	$+.27$	**differ**

Note. Cells report the standardised year coefficient, which is comparable across resources on different native scales; raw slopes are not. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$; p-values are descriptive and uncorrected. The valence and arousal are scored on the same matched collocates within a lexicon, so the sentiment and intensity blocks differ only in which rating is applied.

token set under each lexicon's ratings separates the effect of *which* tokens are scored from the effect of *how* they are scored. Table 5 reports the split.

Coverage does differ, and by more than a rounding error: NRC–VAD matches 866,246 collocate occurrences against Warriner's 801,639, with 87,164 occurrences (10.1% of the NRC total) invisible to Warriner, of which 18.5% are multi-word entries the Warriner norms cannot represent at all. Yet coverage explains little of the disagreement. Averaged over series, the token set accounts for 11% of the movement in the arousal estimate and the ratings for 84%; for ADHD the coverage term is $+0.26$ against a ratings term of -1.11, so restricting NRC–VAD to Warriner's vocabulary leaves the conclusion essentially unchanged while re-reading the same tokens with Warriner's numbers reverses it.

The reason is visible in the resources themselves. On the vocabulary they share, the two lexicons agree closely about valence ($r = .89$ token-weighted, $.83$ type-level)

Table 5
Why the affective lexicons disagree. The shift from the primary to the alternative estimate is split into the part attributable to scoring a different set of tokens and the part attributable to scoring shared tokens differently.

	Standardised β		Decomposition of the shift		
Series	NRC–VAD	Warriner	Total	Coverage	Ratings
Valence (sentiment): NRC–VAD and Warriner ratings correlate $r = .89$ on shared vocabulary					
ADHD	$+.46$	$+.27$	$-.19$	$+.01$	$-.20$
Autism	$+.50$	$+.64$	$+.14$	$+.22$	$-.08$
frustration	$+.67$	$+.64$	$-.02$	$+.05$	$-.08$
loneliness	$+.71$	$+.72$	$+.01$	$+.06$	$-.05$
sadness	$-.79$	$-.84$	$-.05$	$+.09$	$-.14$
Arousal (intensity): NRC–VAD and Warriner ratings correlate $r = .62$ on shared vocabulary					
ADHD	$+.04$	$-.80$	$-.84$	$+.26$	-1.11
Autism	$+.36$	$-.14$	$-.51$	$+.08$	$-.59$
frustration	$-.23$	$-.65$	$-.41$	$-.12$	$-.30$
loneliness	$-.30$	$-.73$	$-.43$	$-.05$	$-.38$
sadness	$+.88$	$+.23$	$-.65$	$-.05$	$-.60$

Note. Both lexicons were applied to the same collocate windows, so their matches can be crossed: the coverage column holds ratings fixed and varies the token set, the ratings column holds the shared token set fixed and varies the rating. Columns sum to the total up to a negligible interaction. Targets are the substantive aggregate; comparators are unframed.

and much less about arousal ($r = .62$ on both). This makes the dimension-specific pattern in Table 4 interpretable rather than mysterious. Valence and arousal are read off *identical* matched collocates within a lexicon, so coverage is held exactly constant between them and the only thing that changes is which rating is consulted. Sentiment survives substitution and intensity does not because the field's two standard affective resources substantially disagree about arousal while broadly agreeing about valence. Any collocate-based intensity measure inherits that disagreement, whatever corpus it is applied to.

5.6.2 Encoders: what the representation is about. Embeddings do not expose which tokens they scored, so no comparable decomposition is available. Two observations stand in its place. First, the two encoders' annual breadth estimates are close to unrelated: across the eleven series they correlate at mean $r = +0.20$, and at $+0.11$ and $+0.17$ for the two overall target series, despite being computed from nearly the same contexts. Whatever year-to-year variation one instrument records, the other largely does not.

Second, we probed what each representation responds to. Taking 3,000 target contexts, we built two minimally edited variants of each: one replacing the target term with the other concept's term, and one replacing an unrelated content word. Table 6 reports how far each encoder moves.

The result is not the one we predicted, and it is more informative. MPNet is not insensitive to the target—it moves slightly further than XL-LEXEME when the target is swapped (.131 against .120). The difference lies in what *else* moves it. Swapping an unrelated content word shifts XL-LEXEME by .0009, effectively nothing, but shifts MPNet by .0106, an order of magnitude more. The ratio of target sensitivity to control sensitivity is $135\times$ for XL-LEXEME against $12\times$ for MPNet.

This has a direct consequence for what breadth measures. Breadth is the dispersion of these representations across a year's contexts. Under XL-LEXEME that dispersion

Table 6
What each breadth encoder responds to. Mean cosine distance between a context and a minimally edited variant of it, over 3,000 target contexts.

Encoder	Target swapped	Control word swapped	Ratio
XL-LEXEME	.1197	.0009	135×
MPNet	.1313	.0106	12×

Note. The target swap replaces the target term with the other concept's term; the control swap replaces an unrelated content word in the same context. Both conditions use the same contexts. A higher ratio means the representation is more specific to the target and less contaminated by the rest of the sentence. Across the 11 annual series the two encoders' breadth estimates correlate at mean $r = .20$.

is almost purely dispersion in target usage, because little else appreciably moves the vector. Under MPNet it also aggregates variation in whatever else the sentences contain—topic, register, boilerplate—so the annual estimate is partly an index of how heterogeneous that year's retained documents were. Two instruments presented as interchangeable operationalisations of contextual breadth are therefore not measuring quite the same construct, and in a corpus whose composition is itself changing they should not be expected to agree.

6. Discussion

6.1 Grading the Evidence

The paper's conclusions do not deserve equal confidence, and the comparison in Section 5.5 is what allows them to be graded. The useful result is that the grading falls along dimensions rather than along findings: one measure is close to invariant and two are not, for a reason that Section 5.6 identifies.

Robust to operationalisation. Neither target broadened horizontally. This is one of the two hypotheses the study set out to test, and no reliable broadening appears under either encoder for either target. Sentiment is likewise close to invariant, agreeing in eight of nine series.

Robust, and independently audited. The compositional shift toward lived-experience framing in ADHD discourse survives an AR(1) check, a period-stratified classifier audit, independent reproduction in the human-adjudicated labels, and an unsupervised distributional check on the frame distinction itself (Section 4.1).

Measurement-conditional. The vertical-creep conclusion depends on the affective lexicon. Under our primary measure intensity is flat; under the framework's original measure ADHD intensity declines reliably. We prefer NRC–VAD on coverage and reliability grounds (Mohammad 2025), but that is an *a priori* argument and does not dissolve the empirical problem. The honest statement is that vertical creep for ADHD cannot presently be adjudicated with these instruments—and, given that the two standard resources correlate at only $r = .62$ on arousal, that limitation is not specific to this study.

Directionally suggestive, not established. The ADHD breadth contraction is the largest effect observed, but breadth is also the least invariant dimension, and the probe in Section 5.6 shows the two encoders are not measuring quite the same thing.

6.2 Composition Is Not Meaning

Frame stratification earned its cost, and the corpus allows the reason to be quantified rather than asserted. Pisl, Bucur, and Podina (2025) and Xiao et al. (2023) warn that a measured affective trend can be produced by a changing mixture of contexts: if clinical contexts are more intense than non-clinical ones, then a corpus drifting toward clinical language will show rising intensity, and one drifting away from it will show falling intensity, whatever happens to the meaning of the term. Both conditions hold here. Clinical contexts are markedly higher in arousal than lived-experience contexts—averaging the first and last year, .025 against .004 for ADHD and .020 against $-.023$ for autism—and the mixture moved: the clinical share of substantive ADHD contexts fell from 72.3% to 63.9% while the lived-experience share rose from 17.4% to 25.7%.

Decomposing the change in the aggregate arousal index into a within-frame and a compositional component makes the consequence explicit. For autism the aggregate fell by .0033 across the window, of which $-.0036$ is compositional and $+.0003$ within-frame: essentially the whole apparent decline is the mixture moving, not the frames. For ADHD the two components have opposite signs—composition $-.0018$, within-frame $+.0035$—and nearly cancel, leaving the flat aggregate the paper reports. Note the sign: the mixture here moved *away* from clinical framing, so what the frame series record is the converse of pathologisation rather than pathologisation itself.

This is the Písl and Xiao mechanism operating in the direction that favours the hypothesis under test. An unstratified analysis of this corpus would have found aggregate arousal drifting downward for autism and would have been entitled, on the usual reasoning, to call it vertical concept creep. The stratified analysis shows the frames themselves did not soften. Our null is therefore conservative: it survives a compositional trend that was pushing the aggregate toward the creep prediction. The same decomposition applies to the valence result, where roughly 40 to 45% of the apparent improvement is compositional rather than a change within either frame.

The composition problem also differs in kind from the one Písl and Xiao address. For *anxiety* or *depression*, stratification must first separate the clinical construct both from ordinary states—everyday anxiety before an exam, feeling depressed after bad news—and from unrelated senses, such as an economic depression or a meteorological one (Xiao et al. 2023). ADHD and autism are diagnostic by definition, so no sense disambiguation is needed, and yet composition still shifts, because what changes is the *framing* of the diagnosis, from disorder construct to identity and lived experience. Composition confounding is therefore not a special problem of polysemous targets; it arises wherever the discourse around a stable referent is itself in motion.

6.3 Semantic Consolidation Rather Than Creep

The substantive pattern is a double null with a directional surprise: ADHD and autism did not drift into milder contexts, did not disperse across a wider range of them, and remained thematically anchored to diagnosis, childhood, and family, while the clearest movement ran toward contraction.

This is intelligible alongside Iacob and Uban's 2026 finding that long-established, codified terms such as *OCD* and *bipolar* moved little on Reddit while newer vernacular terms shifted substantially. ADHD and autism are institutionally anchored: their reference is stabilised by diagnostic manuals, clinical gatekeeping, service categories, and educational law in ways that ordinary-language harm concepts such as *bullying* or *harassment* (Vylomova and Haslam 2021) are not. This suggests a distinction the creep literature has tended to elide. A category can expand *extensionally*, admitting many more people—as ADHD's diagnostic criteria did across successive DSM revisions (Fabiano and Haslam 2020; American Psychiatric Association 2013)—while the typical linguistic contexts of its name, which is all that collocate- and embedding-based measures index, hold their shape. Rising diagnosed prevalence (Zeidan et al. 2022; McKechnie et al. 2023) need not leave the semantic trace a direct reading of concept creep would predict.

What did change is who is talking and in what register. The shift from clinical toward lived-experience framing, much sharper for ADHD, is consistent with the recent growth of adult self-recognition and identity-oriented content (Ginapp et al. 2023; Alper et al. 2025; Martin et al. 2025), and with autism having partly completed that reframing earlier, which would leave its trend flatter—as we find. For the trivialisation concern that motivates much of this literature (Isern-Mas and Almagro 2025; Spencer and Carel 2021), the semantic preconditions are not in evidence in this corpus: the words largely held their shape; the conversation around them did not.

6.4 Implications for LSC Practice

Three recommendations follow, in decreasing order of confidence.

Treat multi-resource comparison as primary, not supplementary. On our evidence a single-resource LSC study reports the properties of a resource as though they were properties of a concept. The remedy is cheap: estimating every trajectory twice cost one additional pass over an existing pipeline, and it converted an unqualified claim about vertical creep into a correctly hedged one. Where two resources disagree, the disagreement is the finding.

Concretely, this suggests a minimal reporting protocol that any study in this tradition could adopt without new data. Estimate each dimension under at least two defensible resources and report both. Compare them on sign, significance, and standardised effect rather than raw magnitude, since resources on different native scales are not otherwise commensurable. Where resources disagree, check whether they disagree about which tokens to score or about how to score them: Section 5.6 shows the two have very different implications, and the check costs one join. Finally, state a conclusion at the strength the weakest concordant resource supports, and mark measurement-conditional findings as such rather than selecting the resource with the cleaner result.

Stratify before estimating. Composition should be modelled explicitly whenever the discourse around a term may be changing, which for socially contested vocabulary is routine. Our classifier is one way to do this; a topic model, a genre classifier, or even recorded metadata strata would serve, and the choice matters less than making one. The general point is that the analyst must decide what population the trend is a trend *in*, and that leaving this implicit does not make the decision go away—it makes it the corpus's decision rather than the analyst's.

Validate the dimensions against human judgement. Neither intensity, breadth, nor connotation as operationalised in this tradition has been benchmarked against human ratings of the same contexts. Until that exists, resource choice rests on plausibility arguments of the kind we ourselves make in Section 4.3, and non-invariance of the kind documented here cannot be adjudicated. This is the most consequential gap our results expose.

6.5 Limitations

The corpus is the most consequential limitation. Common Crawl's coverage is shaped by harmonic centrality, absent social platforms, and growing `robots.txt` blocking over exactly our window (Baack 2024). Findings generalise to quality-filtered English open-web discourse as archived by Common Crawl, not to public discourse at large, and emphatically not to the platforms where much contemporary ADHD and autism discourse now occurs. The final year is also the thinnest, since each crawl captures few pages published in its own year, so 2026 estimates are the least stable in every series.

The frame design inherits classifier error, and the weakest head is the one carrying the headline compositional result; the temporal-stability audit constrains but does not eliminate this. All human labels come from one coder, so no inter-annotator agreement can be reported. The affective measures score lemmatised collocates with context-free lexicon values and are blind to negation, irony, and syntactic scope. The comparators are baselines for negative-affect language, not matched controls in the sense Dubossarsky, Weinshall, and Grossman (2017) recommend. And the design is observational: it characterises how discourse changed, not why.

Finally, the scope of the measurement claim should not be overstated. We demonstrate non-invariance in one well-instrumented case, across two resource substitutions, two targets, three comparators, and three frame strata. That the comparator conclusions flip too makes an idiosyncratic explanation unlikely, but we do not estimate how general the problem is across dimensions, resources, or corpora. Establishing that is the natural next study.

7. Conclusion

We asked whether measured lexical semantic change survives the choices that produce it. Applied to a 13-year web corpus, it depends on the dimension: fifteen of twenty-seven series change their descriptive conclusion when the framework's own alternative lexicon and encoder are substituted, and the reversals reach ordinary emotion vocabulary as readily as diagnostic vocabulary. Sentiment is close to invariant; intensity and breadth are not. Neither failure is mysterious. The two standard affective lexicons agree about valence and disagree about arousal on the very same tokens, so a collocate-based intensity measure inherits a disagreement that has nothing to do with the corpus; and the two standard encoders differ by an order of magnitude in how much non-target material moves them, so their dispersion estimates are not measuring quite the same construct.

The substantive findings are graded accordingly. Neither predicted form of concept creep is supported for ADHD or autism, with the null bounded and, on the shift-share evidence, conservative—the corpus composition was moving in the direction that manufactures apparent vertical creep, and the frames themselves still did not soften. The horizontal result is robust to operationalisation while the vertical one is not, and the clearest change over the period is compositional, a shift from clinical toward lived-experience framing that survives independent audit.

The methodological implication is modest to state and inconvenient to adopt. If a field intends to infer cultural change from lexical measurement, it must first show that the measurement is stable under substitutions its own practitioners regard as innocuous, and diagnose the substitutions that are not. Doing so costs one extra pass over an existing pipeline. Not doing so risks reporting the properties of a lexicon as the properties of a culture.

Table A.1
Period-stratified held-out validation performance for the hierarchical frame classifier. Brackets give 95% Clopper–Pearson intervals; F1 intervals are stratified bootstrap percentiles.

Period	Support	Precision	Recall	FPR	F1
Substantive target discourse					
2014–2017	45/67	.884 [.75, .96]	.844 [.71, .94]	.227 [.08, .45]	.864 [.78, .93]
2018–2022	46/67	.816 [.68, .91]	.870 [.74, .95]	.429 [.22, .66]	.842 [.76, .91]
2023–2026	50/66	.976 [.87, 1.00]	.800 [.66, .90]	.062 [.00, .30]	.879 [.80, .94]
Clinical framing					
2014–2017	36/45	.897 [.76, .97]	.972 [.85, 1.00]	.444 [.14, .79]	.933 [.87, .99]
2018–2022	37/46	.971 [.85, 1.00]	.892 [.75, .97]	.111 [.00, .48]	.930 [.86, .99]
2023–2026	32/50	.833 [.65, .94]	.781 [.60, .91]	.278 [.10, .53]	.806 [.68, .90]
Lived-experience framing					
2014–2017	11/45	.667 [.35, .90]	.727 [.39, .94]	.118 [.03, .27]	.696 [.43, .88]
2018–2022	12/46	.611 [.36, .83]	.917 [.62, 1.00]	.206 [.09, .38]	.733 [.53, .89]
2023–2026	23/50	.792 [.58, .93]	.826 [.61, .95]	.185 [.06, .38]	.809 [.67, .92]

Note. Support gives positive cases over the per-period evaluation denominator. Substantive target discourse is evaluated on all held-out passages in the period; clinical and lived-experience framing are evaluated among human-substantive passages, following the aggregate validation convention. FPR = false-positive rate.

Acknowledgments

I thank Dr Tom Paskhalis for his supervision of the research on which this paper is based.

Appendix A: Classifier Temporal-Stability Audit

Table A.1 reports held-out validation performance for the frozen classifier stratified by period, from the post-hoc audit of the compositional trend in Section 5.1. Periods follow the corpus banding, which balances the 200-passage validation set at 67/67/66. Clinical recall is nominally lower in the late period, but lived-experience error rates are homogeneous across periods and lived-experience precision is highest late—the opposite of the pattern that drifting classifier error would need in order to manufacture the observed rise.

Appendix B: Comparator Trend Models

Table B.1 reports the unframed comparator trend models used to contextualise the target trajectories, across all four measures.

Appendix C: AR(1) Sensitivity Models

Table C.1 reports first-order autoregressive sensitivity estimates for every series flagged for residual autocorrelation in the main tables.

Appendix D: Quadratic Trend Diagnostics

Table D.1 reports quadratic models for the annual series whose linear residuals were flagged. These are diagnostics run because of a residual check, not pre-specified tests, and are reported for completeness rather than as competing trend estimates.

Table B.1
Descriptive annual trend models for baseline comparator trajectories.

Measure	Comparator	$B(SE)$	β	Adj. R^2
Salience	frustration	$-0.0175^{\dagger}$ (0.0167)	-0.30	0.01
Salience	loneliness	0.0056^{*} (0.0021)	0.62	0.33
Salience	sadness	-0.0171^{**} (0.0039)	-0.80	0.61
Sentiment	frustration	$0.0037^{*\dagger}$ (0.0013)	0.67	0.39
Sentiment	loneliness	$0.0044^{**\dagger}$ (0.0013)	0.71	0.45
Sentiment	sadness	-0.0030^{**} (0.0007)	-0.79	0.59
Intensity	frustration	$-0.0004^{\dagger}$ (0.0006)	-0.23	-0.03
Intensity	loneliness	-0.0006 (0.0006)	-0.30	0.01
Intensity	sadness	0.0028^{***} (0.0004)	0.88	0.76
Breadth	frustration	0.0020^{**} (0.0006)	0.72	0.48
Breadth	loneliness	-0.0007 (0.0004)	-0.51	0.19
Breadth	sadness	$-0.0010^{**\dagger}$ (0.0003)	-0.71	0.46

Note. Cells report annual unstandardised OLS slopes as $B(SE)$; β is the standardised year coefficient. Comparator terms are unframed baseline series. Salience uses Common Crawl source year and semantic measures use document publication year. Significance markers: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. † indicates a residual-autocorrelation flag; AR(1) sensitivity estimates for flagged rows are reported in Appendix Table C.1. P values are descriptive and uncorrected.

Table C.1
AR(1) sensitivity models for autocorrelation-flagged annual series.

Source	Measure	Series	Frame	OLS B	AR(1) B	AR(1) p
Table 3	Sentiment	ADHD	Overall	0.0023	0.0037	.147
Table 3	Sentiment	ADHD	Clinical	0.0013	0.0036	.218
Table 3	Sentiment	Autism	Clinical	0.0022	0.0052	.040
Table 3	Breadth	ADHD	Lived experience	-0.0004	-0.0004	.247
Table 3	Breadth	Autism	Overall	0.0001	-0.0001	.753
Table 3	Breadth	Autism	Clinical	0.0007	0.0002	.755
Table 2	Frame classification	Autism	Lived vs clinical	0.59	0.23	.370
Table B.1	Salience	frustration	Baseline	-0.0175	0.0160	.538
Table B.1	Sentiment	frustration	Baseline	0.0037	0.0116	.003
Table B.1	Sentiment	loneliness	Baseline	0.0044	0.0070	.009
Table B.1	Intensity	frustration	Baseline	-0.0004	-0.0022	.078
Table B.1	Breadth	sadness	Baseline	-0.0010	-0.0008	.131

Note. Rows are the series marked with † in the compact regression tables. OLS B repeats the primary annual slope; AR(1) B reports the first-order autoregressive sensitivity slope for the same series. The frame-classification row is reported in percentage points per year; all other rows use the original index units per year. P values are descriptive and uncorrected.

Appendix E: Post-hoc Contributor Diagnostics

Tables E.1–E.3 summarise which collocates carry the valence and arousal estimates, and which content words characterise the highest-distance breadth contexts, grouped into early, middle, and late periods. These are descriptive aids to interpretation, computed after the annual analyses.

Appendix F: Post-hoc Diagnostic on the Frame Distinction

This appendix supports the post-hoc diagnostic reported in Section 4.1. Skip-gram models with 200-dimensional vectors, a context window of 10, a minimum count of 5, and 10 epochs are trained per target-frame corpus and warm-started per publication year, following Vylomova and Haslam (2021). A neighbour is retained only if the canonical

Table D.1
Supplementary quadratic models for autocorrelation-flagged LSC trajectories.

Measure	Target	Frame	Linear $B(SE)$	Quadratic $B_2(SE)$	Δ Adj. R^2	Vertex
Breadth	ADHD	Lived experience	-0.0004 (0.0006)	-0.0000 (0.0002)	-0.11	outside window
Breadth	Autism	Overall	0.0001 (0.0002)	-0.0002^{**} (0.0000)	0.58	Q2 2020 (inverted U)
Breadth	Autism	Clinical	0.0007^{*} (0.0003)	-0.0003^{***} (0.0001)	0.51	Q2 2021 (inverted U)
Sentiment	ADHD	Overall	0.0023 (0.0014)	0.0009^{*} (0.0003)	0.29	Q3 2018 (U)
Sentiment	ADHD	Clinical	0.0013 (0.0015)	0.0012^{**} (0.0003)	0.60	Q2 2019 (U)
Sentiment	Autism	Clinical	0.0022 (0.0015)	0.0013^{***} (0.0002)	0.65	Q1 2019 (U)

Note. Rows are restricted to annual series whose linear-model residuals were flagged for autocorrelation. Linear B is the annual OLS slope from the primary trend model. Quadratic B_2 is the centred-year-squared coefficient; Δ Adj. R^2 is the adjusted-R^2 gain over the linear model. Vertex labels give the calendar quarter containing the fitted decimal-year vertex when it falls inside the observed annual window; otherwise the vertex is marked as outside the window. Significance markers are descriptive and uncorrected: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Table E.1
Post-hoc sentiment collocate contributors by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	Positive: child, adult, add Negative: disorder, autism, depression	Positive: child, like, add Negative: autism, disorder, depression	Positive: child, like, adult Negative: autism, depression, disorder
ADHD	Clinical	Positive: child, adult, add Negative: disorder, depression, autism	Positive: child, like, add Negative: disorder, depression, autism	Positive: child, like, adult Negative: depression, autism, disorder
ADHD	Lived experience	Positive: child, adult, parent Negative: autism, diagnose, dyslexia	Positive: child, like, adult Negative: autism, anxiety, dyslexia	Positive: child, like, adult Negative: autism, dyslexia, diagnose
Autism	Overall	Positive: child, developmental, family Negative: disorder, disability, syndrome	Positive: child, developmental, family Negative: disorder, disability, adhd	Positive: child, family, developmental Negative: disorder, adhd, disability
Autism	Clinical	Positive: child, developmental, include Negative: disorder, disability, disease	Positive: child, developmental, include Negative: disorder, disability, adhd	Positive: child, developmental, include Negative: disorder, adhd, disability
Autism	Lived experience	Positive: child, son, family Negative: disorder, disability, syndrome	Positive: child, son, family Negative: disorder, disability, diagnose	Positive: child, family, son Negative: dangerous, disorder, adhd

Note. Cells show the three largest preprocessed collocate contributors from each direction.

target token occurs at least 20 times in the relevant cell, the neighbour occurs at least five times, it enters an annual top-five list in at least two years, and it has finite estimates in at least ten of the thirteen years. Of 390 annual neighbour rows, 28 neighbours and 364 annual estimates survive.

Figure F.1 shows the contrast Section 4.1 draws on. Within autism the two panels barely overlap: the clinical frame is anchored by *developmental*, *disorder*, *research*, and *study*, the lived-experience frame by *family*, *life*, *people*, and *support*. ADHD separates less sharply, with *disorder*, *symptom*, and *condition* against *help*, which is consistent with the classifier's weaker lived-experience head. The surviving neighbours also stay diagnostic and child- or family-centred across the whole window, but we do not draw that second conclusion here: annual models fitted to yearly slices are known to be unstable (Antoniak and Mimno 2018), so these trajectories are used only to establish that the two frames occupy different distributional neighbourhoods, not to estimate how those neighbourhoods changed over time.

Table E.2
Post-hoc arousal collocate contributors by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	Raising: disorder, anxiety, hyperactivity Lowering: add, treatment, patient	Raising: anxiety, disorder, hyperactivity Lowering: add, sleep, student	Raising: anxiety, disorder, challenge Lowering: individual, common, treatment
ADHD	Clinical	Raising: disorder, anxiety, hyperactivity Lowering: treatment, add, patient	Raising: disorder, anxiety, hyperactivity Lowering: add, sleep, narcolepsy	Raising: anxiety, disorder, hyperactivity Lowering: treatment, common, individual
ADHD	Lived experience	Raising: anxiety, suffer, struggle Lowering: parent, student, add	Raising: anxiety, challenge, disorder Lowering: student, add, dyslexia	Raising: challenge, anxiety, struggle Lowering: autistic, dyslexia, individual
Autism	Overall	Raising: disorder, developmental, spectrum Lowering: link, family, individual	Raising: disorder, developmental, spectrum Lowering: people, individual, family	Raising: disorder, developmental, dangerous Lowering: individual, people, family
Autism	Clinical	Raising: disorder, developmental, schizophrenia Lowering: link, center, individual	Raising: disorder, developmental, spectrum Lowering: link, individual, center	Raising: disorder, developmental, schizophrenia Lowering: individual, therapy, link
Autism	Lived experience	Raising: disorder, boy, spectrum Lowering: son, family, people	Raising: disorder, challenge, spectrum Lowering: people, son, individual	Raising: dangerous, obsession, disorder Lowering: individual, people, family

Note. Cells show the three largest preprocessed collocate contributors from each direction.

Table E.3
Post-hoc breadth high-distance content words by target, frame, and period.

Target	Frame	2014-2017	2018-2021	2022-2026
ADHD	Overall	disorder, behavior, childhood, impulsivity, inattention	disorder, include, child, symptom, syndrome	disorder, child, autism, learning, learn
ADHD	Clinical	disorder, behavior, childhood, impulsivity, inattention	disorder, include, syndrome, autism, disability	disorder, condition, autism, use, learning
ADHD	Lived experience	disorder, thing, child, parent, school	disorder, autism, student, diagnose, people	disorder, autism, dyslexia, spectrum, autistic
Autism	Overall	lead, need, time, society, channel	world, day, feel, high, thing	lead, research, fact, know, actually
Autism	Clinical	family, individual, research, speaks, government	child, early, new, organization, speaks	speaks, need, developmental, treatment, research
Autism	Lived experience	far, car, actually, attitude, away	world, day, feel, high, work	experience, think, people, program, speaks

Note. Cells show the five most frequent filtered content lemmas among the twenty highest-distance contexts in each cell. This breadth diagnostic summarises high-distance context wording rather than VAD collocates.

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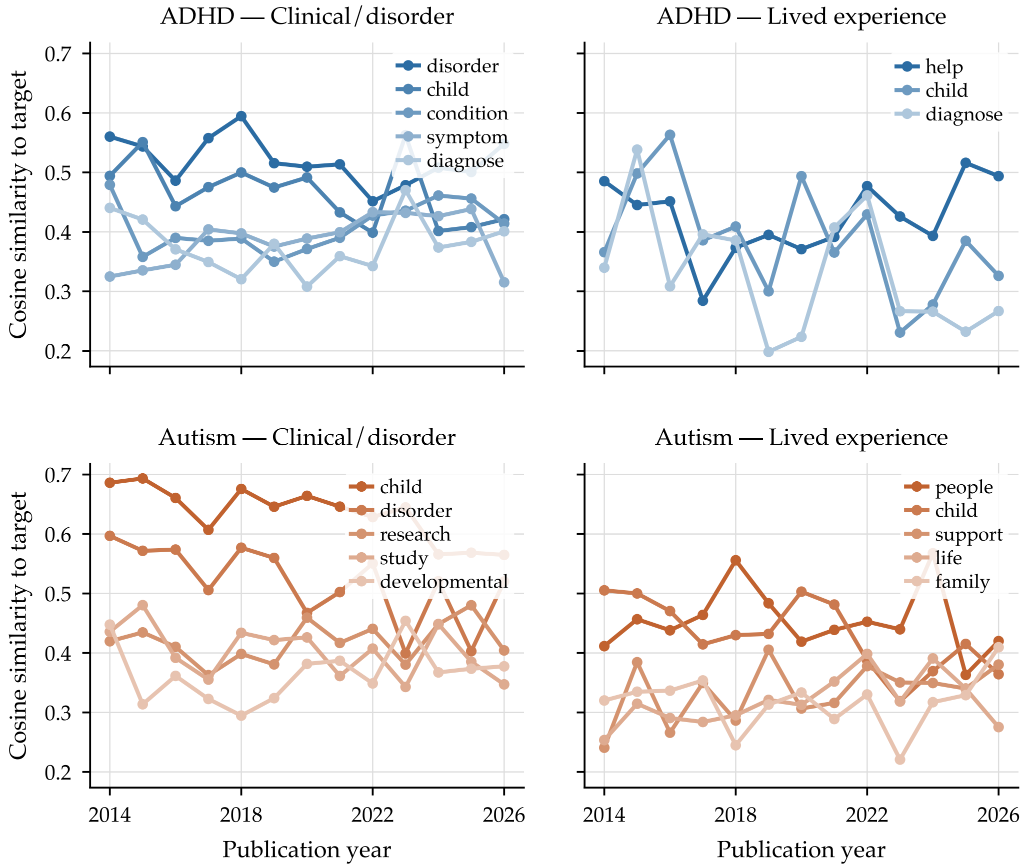

Figure F.1
Annual cosine similarity between each target concept token and its stable distributional neighbours, within the clinical and lived-experience frames. Neighbours are ordered in each legend by mean similarity. The overall stratum is omitted: pooling the frames answers a different question from the one this diagnostic asks.

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